

ICT 1209

Web Technologies

1.Introduction

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Course Content and Objectives

- Delivered in 15 weeks, covering various topics, such as World wide web, HTML, CSS, Client-Side Scripting, Server-Side Scripting, Session handling, Cookies, Version Controlling, and New trends in web technologies.
- Learning outcomes:
 1. Understanding the development of a client-side browser-based web application.
 2. Demonstrate web design techniques.
 3. Create functioning web applications.
 4. Reflect on and evaluate the appropriateness of the chosen element, technologies and architectures.

Assessment Structure

- Continuous Assessments : 40%
 - Mini Project - 20%
 - 02 In-class Tests - 5% each (dates will be announced 2 weeks before)
 - Practical Evaluation - 10%
- End Semester Examination (Theory) : 60%

Announcements

Enroll in LMS Course: <https://techlms.rjt.ac.lk/course/view.php?id=36>

Enrollment Key : **ICT1209_2024**

Module Completion

The requirement of achieving 40% from Assignment marks and 35% from End Examination marks.

Class Policies

- No unnoticed late arrivals.
- Following instructions, originality will receive marks in evaluations.
- Deadlines are deadlines. Start early. Work regularly. Deadlines are checkpoints, not starting signals.

**a student at rest will remain at rest unless
acted upon by deadline**



Resources

- W3Schools Online Web Tutorials
- “Internet & World Wide Web How to Program”, Fifth Edition. Paul J. Deitel, Harvey M. Deitel, Abbey Deitel, Pearson, 2011
- "Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics" Jennifer Niederst Robbins

Introduction to WWW

What is WWW?

- The World Wide Web (WWW) is an information space that allows sharing of data and documents across the Internet.
- The Web was invented in 1989 by Tim Berners-Lee while he was working at CERN, the European Organization for Nuclear Research in Switzerland.
- Berners-Lee envisioned the Web as a tool for sharing scientific information among researchers, but it quickly grew to become a platform for communication, commerce, and entertainment around the world.

Keywords

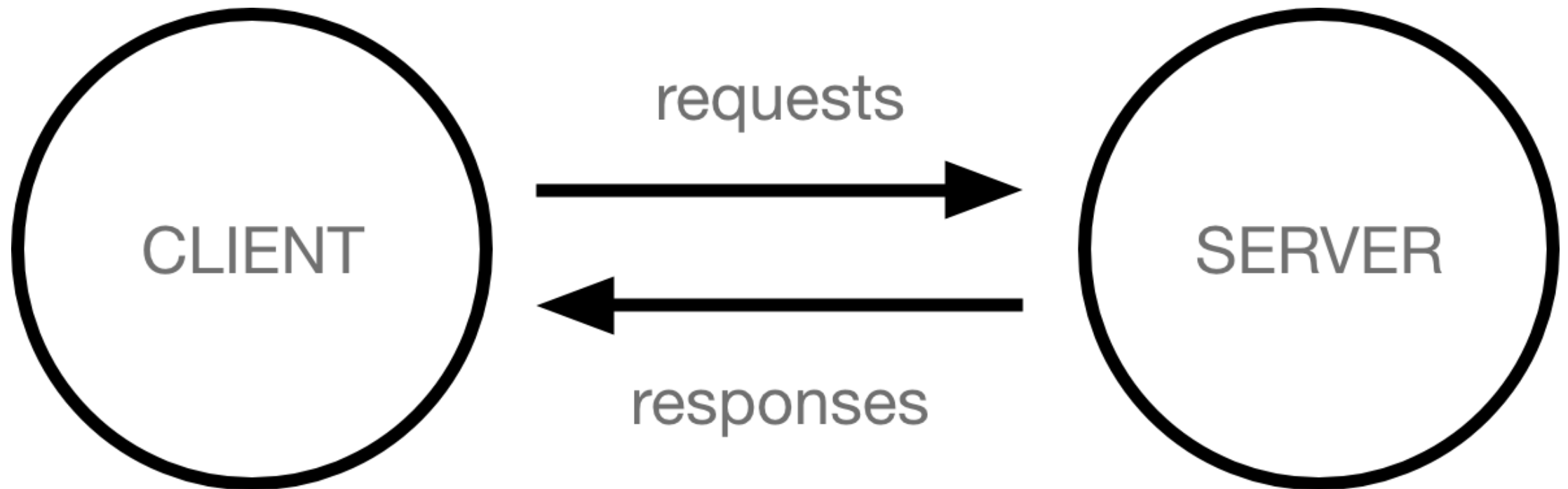
WWW and Web Sites: a website is a specific entity that exists on the Web and contains a set of web pages and other resources, while the Web is the larger network that connects all these resources together.

Web Server : A web server is a software application that runs on a computer and is responsible for serving content, such as web pages, to clients over the World Wide Web

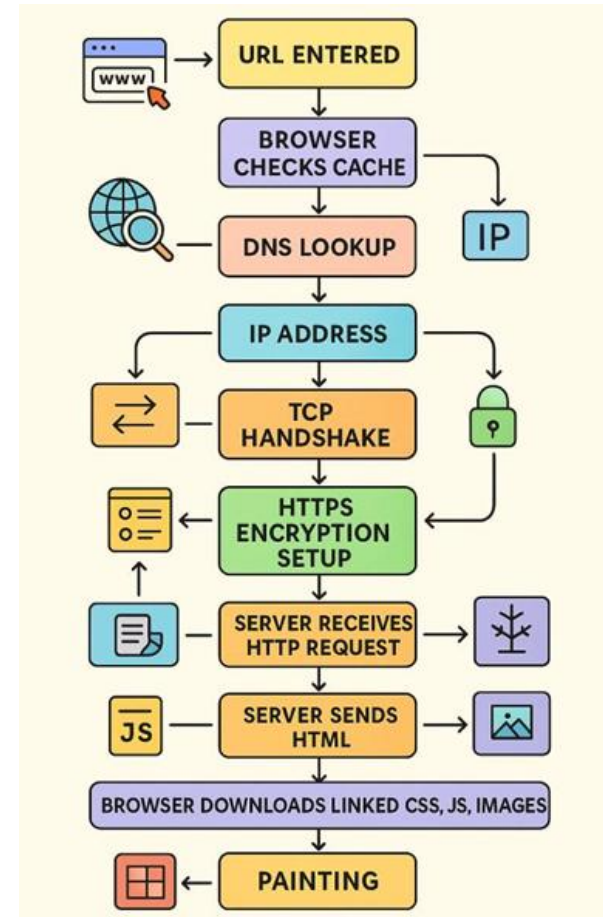
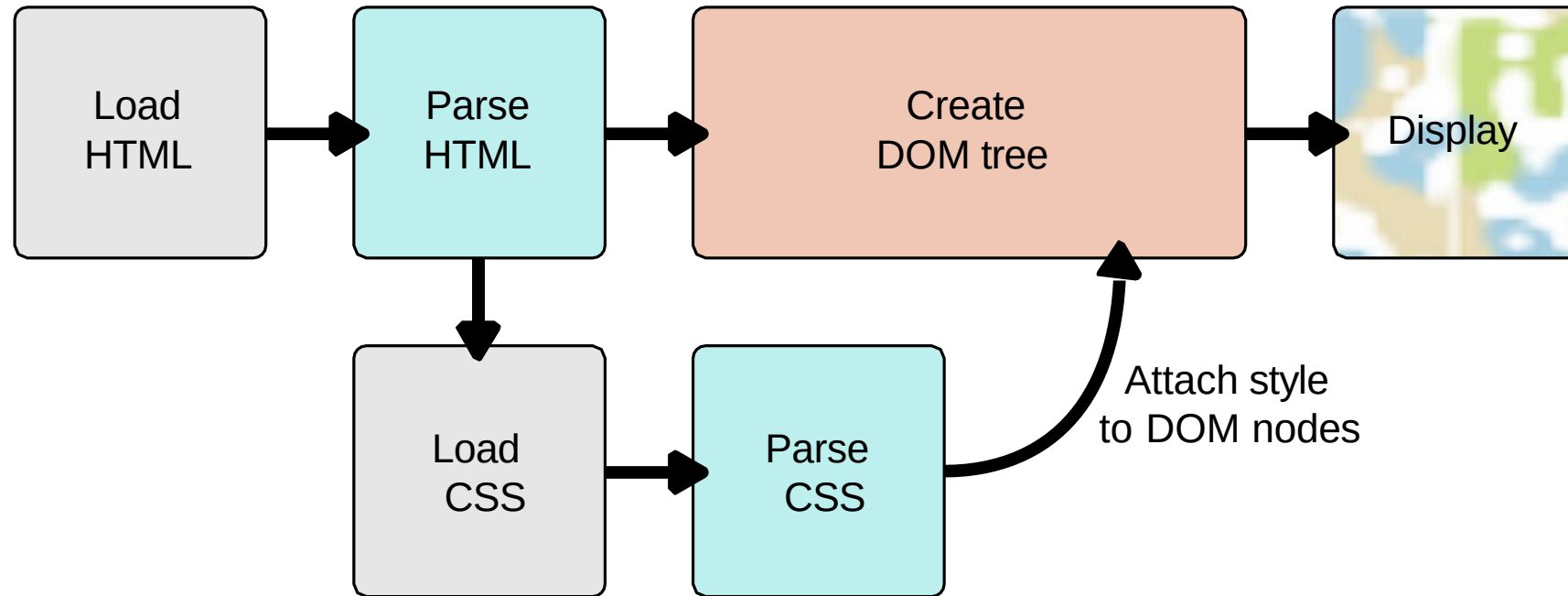
Web Browser : A software application that is used to access and view content on the [WWW](http://www.). provide users with a graphical user interface (GUI) for accessing and navigating web content.

Search Engine : A software program that allows users to search for information on the World Wide Web. It enables users to enter keywords or search terms and then displays a list of relevant web pages, documents, images, and other types of content that match those search terms.

How does a Web Page load?



User Enters the URL → DNS translation → Connect to Server (SYN- ACK)
And,

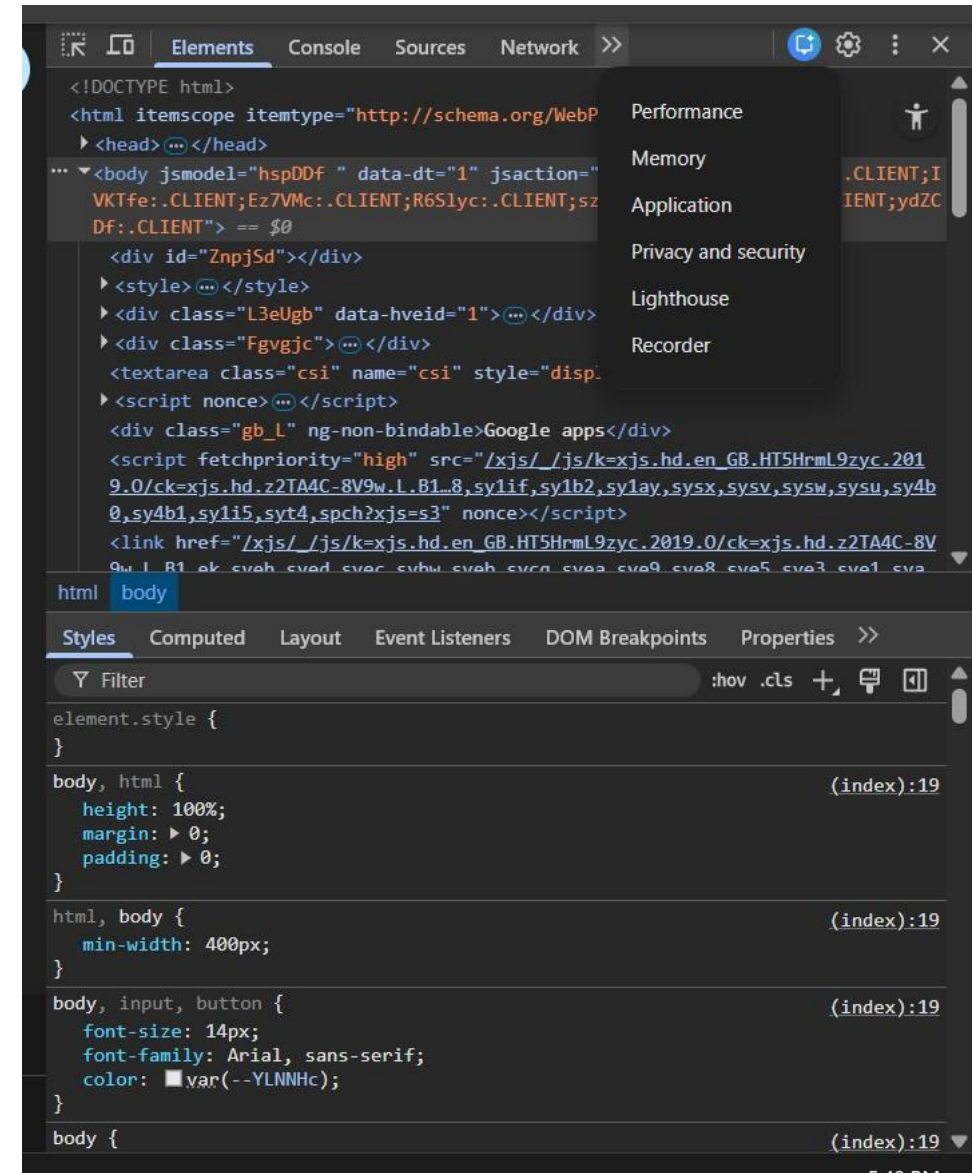


Introducing Browser Developer Tools

This is where we see real browser–server communication

Dev Tools

- DevTools are a set of web development tools built into the Web browser that allow you to edit pages, diagnose problems, and monitor site performance.
- You can access them using the keyboard shortcut *Ctrl+Shift+I* (Windows/Linux)



HTTP Status Codes

- A set of standardized and agreed upon numbers that identify information about an http response.
- Status codes are 3 digits and broken down into 5 categories, each organized by the first digit of the number.
- 1xx: **Informational** provides information about a request in progress.
- 2xx: **Successful** indicates that the request was received and accepted.
- 3xx: **Redirection** indicates that the request has to be redirected.
- 4xx: **Client Error** indicates that an error has occurred on the client.
- 5xx: **Server Error** indicates that an error has occurred on the server.

Network

Performance

Memory

Application

Preserve log

Disable cache

No throttling

Filter

Invert

More filters

All

Fetch/XHR

Doc

CSS

JS

Font

Img

Media

Manifest

Socket

Wasm

Other

1,000 ms

2,000 ms

3,000 ms

4,000 ms

5,000 ms

6,000 ms

7,000 ms

8,000 ms

9,000 ms

10,000 ms

11,000 ms

12,000 ms

13,000 ms

14,000 ms

15,000 ms

Name	Status	Type	Initiator	Size	Time	Fulfilled by
shop_06-1-1.jpg	404	text/html	(index):3762	81.2 kB	1.16 s	
shop_04-1-1.jpg	404	text/html	(index):3762	81.2 kB	1.22 s	
shop_08-1-1.jpg	404	text/html	(index):3762	49.4 kB	1.16 s	
swiper.min.js?ver=8.4.5	200	script	frontend.min.js?ver=3.25.10:2	37.6 kB	286 ms	
shop_01-2-1.jpg	404	text/html	(index):3762	81.2 kB	961 ms	
con-2024.jpg	200	jpeg	rs6.min.js?ver=6.5.5:36	743 kB	690 ms	
sleder-1.png	200	png	rs6.min.js?ver=6.5.5:36	104 kB	290 ms	
shop_03-2-1.jpg	301	text/html / Red...	swiper.min.js?ver=8.4.5:13	61 B	668 ms	
Postgraduate-Programmes-150x150.png	200	png	swiper.min.js?ver=8.4.5:13	29.1 kB	515 ms	
Postgraduate-Diploma-in-Rural-Development-Sinhala-1...	200	png	swiper.min.js?ver=8.4.5:13	33.9 kB	526 ms	
Postgraduate-Diploma-in-Rural-Development-150x150....	200	png	swiper.min.js?ver=8.4.5:13	33.5 kB	533 ms	
Master-of-Agriculture-MAgri-150x150.png	200	png	swiper.min.js?ver=8.4.5:13	32.1 kB	578 ms	
Diploma-in-Business-Economics-150x150.jpeg	200	jpeg	swiper.min.js?ver=8.4.5:13	7.1 kB	577 ms	
Diploma-in-Agrpreneurship-150x150.jpeg	200	jpeg	swiper.min.js?ver=8.4.5:13	8.8 kB	580 ms	

69 requests 6.6 MB transferred 9.8 MB resources

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Guess what the code means

Code : 200



- A. The server accepted the request, but it is **not ready** to return the result yet.
- B. The request was **successful**, and the server is returning the requested resource.
- C. The resource has been **permanently moved** to a new URL.
- D. The server understood the request but **refuses to authorize** it.

Code : 204



- A. A. The server couldn't understand the request due to invalid syntax.
- B. The request was successful, but the server has no content to return.
- C. The resource could not be found at the given URL.
- D. The server encountered an internal unexpected error.

Code : 404



- A. The server understood the request but requires authentication.
- B. The request was successful, but there is no content to return.
- C. The server cannot find the requested resource at the given URL.
- D. The server is temporarily unable to handle the request due to overload.

Code: 503



- A. A. The server successfully processed the request but has no content to return.
- B. The server is temporarily unable to handle the request due to overload or maintenance.
- C. The requested resource has been permanently moved to a new location.
- D. The request was malformed and the server could not understand it.

URL - Uniform Resource Locator

- A string of characters that is used to identify and locate resources on the World Wide Web. A URL typically consists of several parts, including the protocol, the domain name, the path, and any parameters or queries.

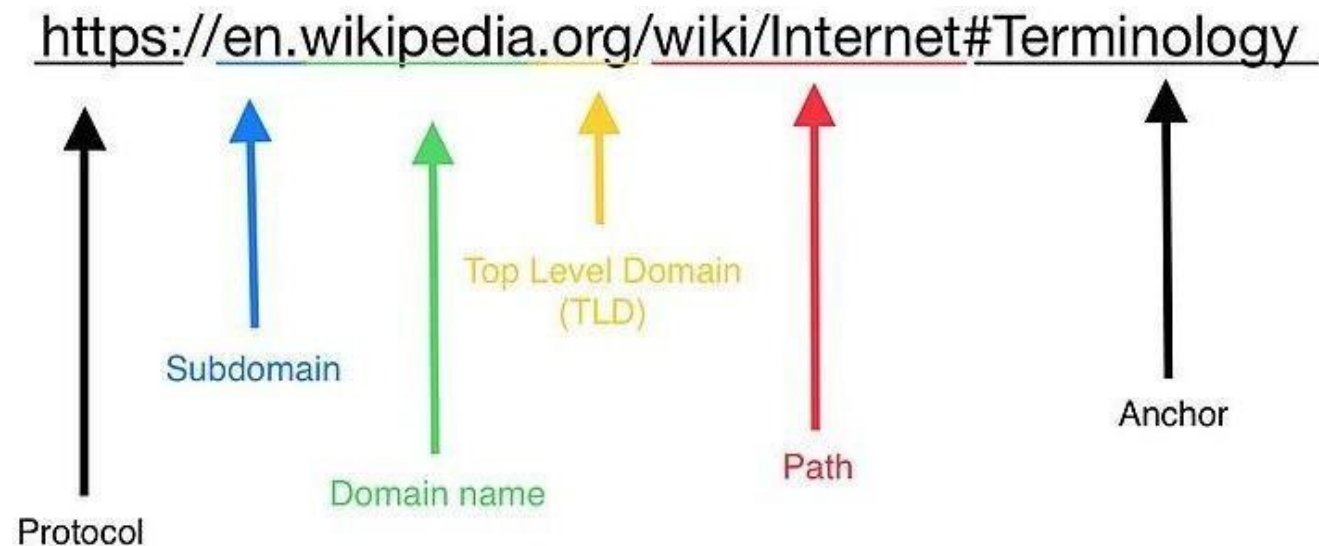
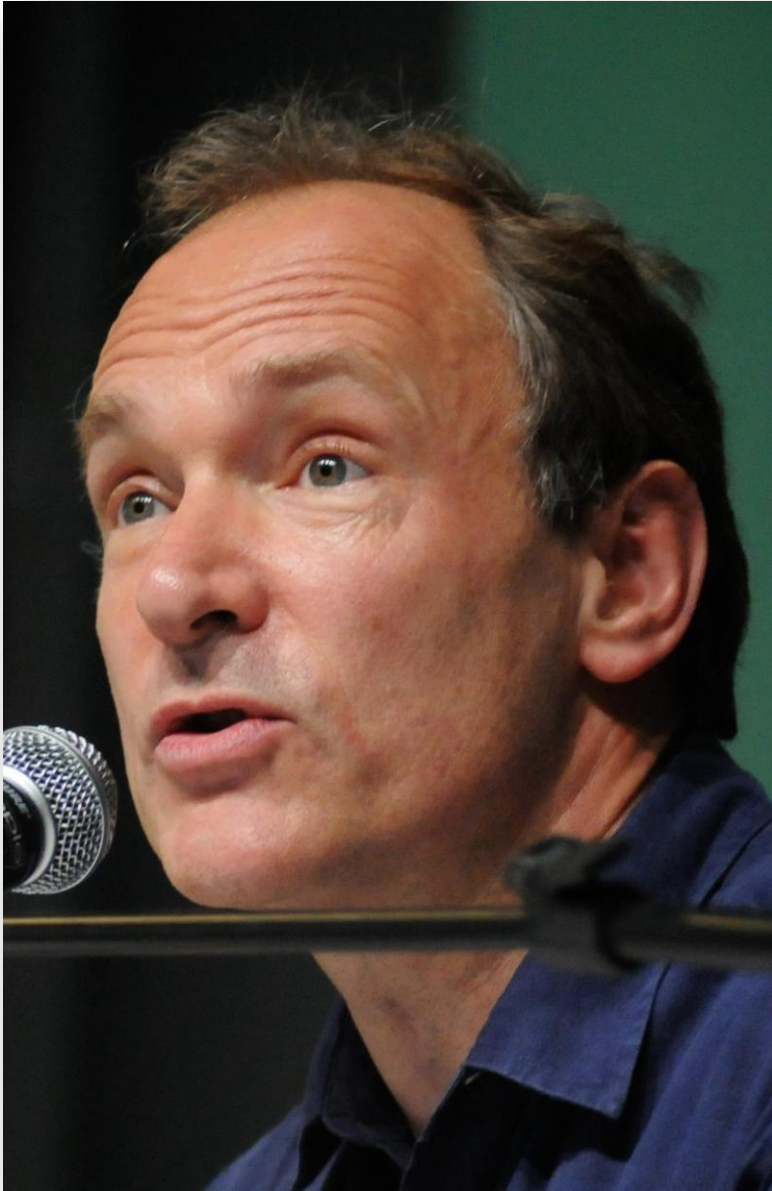


Image Source : https://commons.wikimedia.org/wiki/File:URL_structure.jpg

- The protocol is the method used to access the resource, such as "http" or "https".
- The domain name is the address of the server that hosts the resource, and may include the top-level domain (TLD) like .com, .org, .net, or country code TLDs like .uk, .de, or .fr.
- The path specifies the location of the resource on the server, and can include directories and file names. Parameters or queries are optional, and allow users to provide additional information to the server, such as search terms or user preferences.

Eg: URL for the Google homepage is "<https://www.google.com>". In this case, "https" is the protocol, "www.google.com" is the domain name, and there is no path or parameters.

Bit of History



- In 1991, Berners-Lee published the first website, which provided information about the Web and how to use it.
- He also created the first Web browser, which was also called WorldWideWeb (Nexus).
- The first website contained basic information about the Web, including instructions on how to use a browser and create a Web page.

Tim_Berners-Lee_CP_2.jpg: Silvio Tanaka derivative work: AnOddName (talk), Tim_Berners-Lee_CP_2.jpg / CC BY 2.0

Zdroj: <https://citaty.net/autori/tim-berners-lee/>

The 1st ever Website

World Wide Web

The WorldWideWeb (W3) is a wide-area [hypermedia](#) information retrieval initiative aiming to give universal access to a large universe of documents.

Everything there is online about W3 is linked directly or indirectly to this document, including an [executive summary](#) of the project, [Mailing lists](#) , [Policy](#) ,

[What's out there?](#)

Pointers to the world's online information, [subjects](#) , [W3 servers](#), etc.

[Help](#)

on the browser you are using

[Software Products](#)

A list of W3 project components and their current state. (e.g. [Line Mode](#) , [X11 Viola](#) , [NeXTStep](#) , [Servers](#) , [Tools](#) , [Mail robot](#) , [Library](#))

[Technical](#)

Details of protocols, formats, program internals etc

[Bibliography](#)

Paper documentation on W3 and references.

[People](#)

A list of some people involved in the project.

[History](#)

A summary of the history of the project.

[How can I help ?](#)

If you would like to support the web..

[Getting code](#)

Getting the code by [anonymous FTP](#) , etc.

Modern Web Apps

- Mobile friendly design
- Responsive and adaptive layouts
 - Websites use fixed layouts for specific screen sizes
- Progressive Web Apps (PWAs)
 - Web apps that behave like mobile apps with offline support and installability.
- Single Page Applications (SPAs)
 - A web app that loads one HTML page and updates content dynamically without full reloads.



Static and Dynamic Web Pages

Aspect	Static Pages	Dynamic Pages
Content	Fixed content that does not change unless manually updated. Same for all users	Content changes based on user input, preferences, or server-side logic.
Speed	Faster to load because no server processing is needed.	Slightly slower because server-side processing generates content.
Maintenance	Easier to maintain; no backend logic or dynamic generation required.	More complex to maintain due to scripts, databases, and server-side processing.
Interactivity	Limited interactivity.	Highly interactive – supports forms, search, personalized content, social media features, etc.

Web 1.0, 2.0 and 3.0

Web 1.0	Web 2.0	Web 3.0
• Early days of the WWW • Static web pages providing information • One-way flow of information (read-only) • Simple pages with text and static images	• Current stage of the WWW • High interactivity and user participation • Users create, share, and interact with content • Includes social media, wikis, blogs, and collaborative apps	• Also called the Semantic Web • Next generation of the WWW • Aims to make the web intelligent and machine-understandable

Note: Web 3.0 and Web3 are two different concepts.

Is Web and Internet the same?

Internet is the **infrastructure** that allows for the exchange of information and communication, while the World Wide Web is **one of many applications** that use the Internet to provide access to information and other resources

- The Internet is a global network of computers and other devices connected together, enabling the exchange of information and communication across vast distances. The Internet allows for a wide range of services and applications, including email, instant messaging, file sharing, online gaming, and more.
- The World Wide Web, on the other hand, is a collection of interconnected documents and other resources, linked by hyperlinks and URLs, that are accessed over the Internet. The Web provides a platform for publishing and accessing information, including text, images, audio, and video.

Protocols governing the web

1. **Hypertext Transfer Protocol (HTTP):** HTTP is the protocol used for transferring data over the web. It defines the format of the request and response messages that are exchanged between the client and the server.
2. **Transmission Control Protocol (TCP):** TCP is a protocol that provides reliable, ordered, and error-checked delivery of data between applications running on different hosts. It works together with **IP (Internet Protocol)**, which provides the routing mechanism to ensure that data is delivered to the correct destination.
3. **Domain Name System (DNS):** DNS is a protocol that translates domain names into IP addresses. It allows users to access websites using a memorable name (such as www.example.com) instead of a numerical IP address (such as 192.168.1.1).
4. **Secure Sockets Layer (SSL) and Transport Layer Security (TLS):** SSL and TLS are protocols used for secure communication over the web. They provide encryption and authentication to ensure that sensitive data is transmitted securely.

World Wide Web Consortium (W3C)

- An international community that develops standards for the web. It was founded in 1994 by Tim Berners-Lee, the inventor of the World Wide Web. The aim of W3C is to promote the growth of the web and ensure that it remains accessible to everyone, regardless of their technology or ability.
- W3C is responsible for developing and maintaining [a wide range of web standards](#), including HTML, CSS, and JavaScript. It also develops standards for web accessibility, mobile web, and internationalization, among other areas

Web Programming Technologies

- Front-end languages: HTML, CSS, JavaScript
- Back-end languages: PHP, python, Java
- Front-end frameworks and libraries: Angular.js, React.js, jQuery, SASS, and Flutter.
- Back-end frameworks and libraries: Laravel, Express, Spring, and Django.
- Full Stack development: This includes front-end and back-end as well as databases

Thank you

Next : HTML